Matrice di test — syscall throttling (SOA 2025-26)

Gli screenshot sono stati acquisiti quando i programmi di test risiedevano in `user/`; nella struttura attuale del repository sono in `test/` (il comando diventa `./test/test_...`). `throttlectl` resta in `user/`.

Initial setup

```
# discoverer caricato
ls /sys/module/the_usctm || sudo insmod ~/esempi/Linux-sys_call_table-discoverer/the_usctm.ko
cd ~/syscall-throttling-SOA_25-26
make clean && make
```

A ogni aggiornamento di kernel sia `the_usctm` sia `throttler` vanno ricompilati.

```
cd ~/esempi/Linux-sys_call_table-discoverer
make clean && make
modinfo the_usctm.ko | grep vermagic
sudo insmod the_usctm.ko

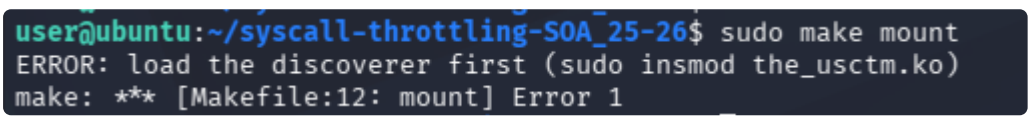
cd ~/syscall-throttling-SOA_25-26
make clean && make
sudo make mount
lsmod | grep throttler
```

A — Ciclo di vita del modulo

A0 — Dipendenza dal discoverer

Con `the_usctm` non caricato, `insmod` deve fallire in modo pulito (indirizzo tabella a 0 → `init_module` ritorna errore, messaggio in `dmesg`).

Atteso: nessun crash.



A1 — Caricamento

`sudo make mount` → `lsmod | grep throttler` presente; `dmesg` mostra device registrato.

Atteso: nessun errore, device `/dev/throttler` creato.

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo insmod ~/esempi/Linux-sys_call_table-discoverer/the_usctm.ko
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo make mount
insmod throttler.ko the_syscall_table=18446744072564507424
mknod /dev/throttler c $(grep throttler /proc/devices | cut -d' ' -f1) 0
user@ubuntu:~/syscall-throttling-SOA_25-26$ lsmod | grep throttler
throttler              28672  0
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo dmesg | tail -5
[  318.003249] USCTM: found sys_ni_syscall entry at syscall_table[212]
[  318.003250] USCTM: found sys_ni_syscall entry at syscall_table[214]
[  318.016640] USCTM: a sys_call with 2 parameters has been installed as a trial on the sys_call_table at displacement 134
[  318.016644] USCTM: module correctly mounted
[  321.286331] THROTTLER: new device registered, it is assigned major number 241
```

A2 — Scaricamento semplice

Nessun thread in gioco, `sudo make unmount` → modulo rimosso, tabella ripristinata (`dmesg`: `syscall ... restored`, device unregistered).

Atteso: nessun crash/hang.

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo make unmount
rm -f /dev/throttler
rmmod throttler
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo dmesg | tail -5
[  318.003250] USCTM: found sys_ni_syscall entry at syscall_table[214]
[  318.016640] USCTM: a sys_call with 2 parameters has been installed as a trial on the sys_call_table at displacement 134
[  318.016644] USCTM: module correctly mounted
[  321.286331] THROTTLER: new device registered, it is assigned major number 241
[  404.199527] THROTTLER: new device unregistered, it was assigned major number 241
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo make unmount
rm -f /dev/throttler
rmmod throttler
rmmod: ERROR: Module throttler is not currently loaded
make: *** [Makefile:18: unmount] Error 1
```

A3 — insmod/rmmod ripetuti

5 cicli `unmount` / `mount` di fila.

Atteso: nessun leak, nessun warning in `dmesg`, ogni ciclo pulito.

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo make mount
insmod throttler.ko the_syscall_table=18446744072564507424
mknod /dev/throttler c $(grep throttler /proc/devices | cut -d' ' -f1) 0
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo make unmount
rm -f /dev/throttler
rmmod throttler
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo make mount
insmod throttler.ko the_syscall_table=18446744072564507424
mknod /dev/throttler c $(grep throttler /proc/devices | cut -d' ' -f1) 0
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo make unmount
rm -f /dev/throttler
rmmod throttler
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo make mount
insmod throttler.ko the_syscall_table=18446744072564507424
mknod /dev/throttler c $(grep throttler /proc/devices | cut -d' ' -f1) 0
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo make unmount
rm -f /dev/throttler
rmmod throttler
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo make mount
insmod throttler.ko the_syscall_table=18446744072564507424
mknod /dev/throttler c $(grep throttler /proc/devices | cut -d' ' -f1) 0
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo make unmount
rm -f /dev/throttler
rmmod throttler
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo make mount
insmod throttler.ko the_syscall_table=18446744072564507424
mknod /dev/throttler c $(grep throttler /proc/devices | cut -d' ' -f1) 0
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo make unmount
rm -f /dev/throttler
rmmod throttler
```

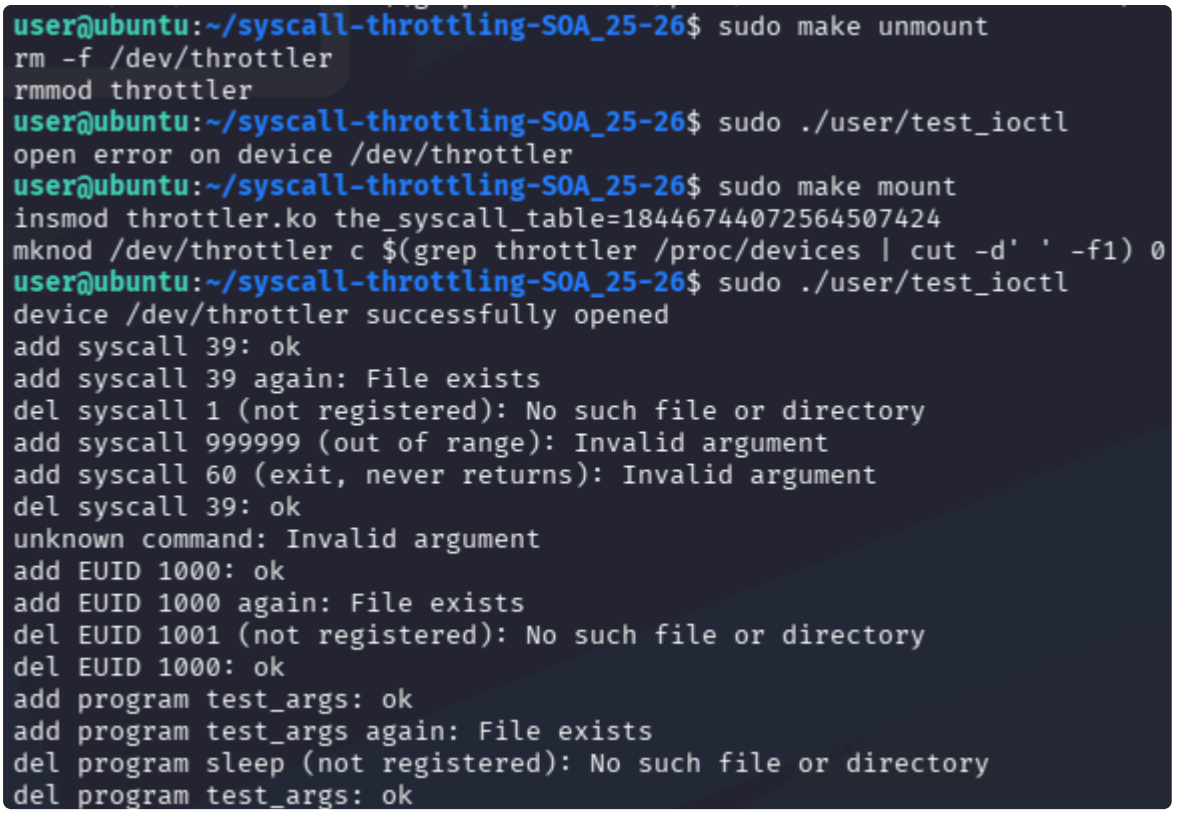
B — Registrazione e percorsi d'errore (con sudo)

Setting: `sudo make unmount && sudo make mount`

B1 — Tutti i comandi + casi d'errore

`sudo ./test_ioctl`
add/del syscall*
add/del euid`
add/del prog
Doppia registrazione
Deregistrazione di non registrato
*Syscall fuori intervallo***
***Syscall che non ritornano*

Atteso: il file `test_ioctl` indica già su stdout l'esito atteso.



B8 — Registro pieno

Registra 20 syscall valide distinte `sudo ./user/throttlectl add-sys #nr`

Atteso: alla 21ª `-ENOSPC` ("No space left on device").

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 1
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 2
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 3
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 4
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 5
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 6
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 7
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 8
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 9
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 10
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 11
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 12
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 13
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 14
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 15
add-sys: Invalid argument
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 16
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 17
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 18
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 19
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 20
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 21
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 22
add-sys: No space left on device
```

B9 — Nome programma troncato

`sudo ./user/throttlectl add-prog nome_molto_lungo_oltre_15`

Atteso: registrato troncato a 15 caratteri; `show` mostra la versione troncata. (Coerente con `TASK_COMM_LEN` =16)

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-prog nome_molto_lungo_oltre_15
add-prog: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl show
monitor: off
MAX: 0
syscall registered: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 16 17 18 19 20 21
EUID registered:
registered programs: nome_molto_lung
```

C — Permessi

C1 — Lettura aperta a tutti

`./user/throttlectl show` **senza** sudo

Atteso: funziona (GET_CONFIG gestito prima del controllo EUID).

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl show
monitor: off
MAX: 0
syscall registered: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 16 17 18 19 20 21
EUID registered:
registered programs: nome_molto_lung
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl show
monitor: off
MAX: 0
syscall registered: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 16 17 18 19 20 21
EUID registered:
registered programs: nome_molto_lung
```

C2 — Scrittura negata ai non root

Senza sudo: `add-sys 39`, `set-max 5`, `on`, `off`, `add-prog x`

Atteso: tutte `-EPERM` ("Operation not permitted").

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl add-sys 1
add-sys: Operation not permitted
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl add-sys 2
add-sys: Operation not permitted
```

C3 — Scrittura consentita a root

Con `sudo`: gli stessi comandi

Atteso: ok.

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 1
add-sys: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl add-sys 2
add-sys: ok
```

D — Filtro: chi viene osservato

Setting: `sudo make unmount && sudo make mount && sudo ./user/throttlectl on && sudo ./user/throttlectl add-sys 39`

D1 — Match per programma

`sudo ./user/throttlectl add-prog test_syscall_by_nr` poi `./test_syscall_by_nr 39 1`

Atteso: `dmesg` : `observed - program test_syscall_by` (nome troncato).

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl show
monitor: on
MAX: 10
syscall registered: 39
EUID registered:
registered programs: test_syscall_by
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/test_syscall_by_nr 39 1
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo dmesg | tail -2
[ 1886.902510] THROTTLER: device file closed
[ 1893.205657] THROTTLER: syscall 39 observed - program test_syscall_by, thread 1635
```

D2 — Match per euid

`sudo ./user/throttlectl del-prog test_syscall_by_nr; sudo ./user/throttlectl add-uid 1000` poi `./test_syscall_by_nr 39 1`

Atteso: `dmesg` : `observed - euid 1000` .

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl show
monitor: on
default MAX: unlimited
syscall registered: 39
EUID registered: 1000
registered programs:
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/test_syscall_by_nr 39 1
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo dmesg | tail -10
[ 2915.392427] THROTTLER: EUID 1000 registered
[ 2915.392863] THROTTLER: device file closed
[ 2920.394196] THROTTLER: syscall 39 observed - euid 1000, thread 2189
[ 2920.397509] THROTTLER: device file successfully opened for object with minor 0
[ 2920.397528] THROTTLER: somebody called an ioctl on dev with [major,minor] number [241,0], command 2183164938 and paramet
er 140734861549712
[ 2920.399954] THROTTLER: device file closed
[ 2924.777770] THROTTLER: syscall 39 observed - euid 1000, thread 2190
[ 2924.781542] THROTTLER: syscall 39 observed - euid 1000, thread 2190
[ 2928.644553] THROTTLER: syscall 39 observed - euid 1000, thread 2191
[ 2928.645400] THROTTLER: syscall 39 observed - euid 1000, thread 2192
```

D3 — Match doppio (prog + euid)

`sudo ./user/throttlectl add-prog test_syscall_by_nr` poi `./test_syscall_by_nr 39 1`

Atteso: `dmesg` : `observed - euid 1000 and program test_syscall_by` (una sola riga per chiamata).

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl show
monitor: on
default MAX: unlimited
syscall registered: 39
EUID registered: 1000
registered programs: test_syscall_by
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/test_syscall_by_nr 39 1
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo dmesg | tail -10
[ 2969.524894] THROTTLER: program test_syscall_by registered
[ 2969.525217] THROTTLER: device file closed
[ 2980.380304] THROTTLER: syscall 39 observed - euid 1000, thread 2199
[ 2980.383989] THROTTLER: device file successfully opened for object with minor 0
[ 2980.384010] THROTTLER: somebody called an ioctl on dev with [major,minor] number [241,0], command 2183164938 and paramet
er 140731628683456
[ 2980.384796] THROTTLER: device file closed
[ 2989.472789] THROTTLER: syscall 39 observed - euid 1000, thread 2203
[ 2989.477104] THROTTLER: syscall 39 observed - euid 1000 and program test_syscall_by, thread 2203
[ 3001.376282] THROTTLER: syscall 39 observed - euid 1000, thread 2204
[ 3001.377199] THROTTLER: syscall 39 observed - euid 1000, thread 2205
```

D4 — Nessun match

`sudo ./user/throttlectl del-uid 1000; sudo ./user/throttlectl del-prog test_syscall_by_nr` poi `./test_syscall_by_nr 39 1`

Atteso: **nessuna** riga `observed`.

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl del-uid 1000; sudo ./user/throttlectl del-prog test_sys
call_by_nr
del-uid: ok
del-prog: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl show
monitor: on
default MAX: unlimited
syscall registered: 39
EUID registered:
registered programs:
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo dmesg | tail -10
[ 3259.811325] THROTTLER: somebody called an ioctl on dev with [major,minor] number [241,0], command 29703 and parameter 10
00
[ 3259.811335] THROTTLER: EUID 1000 deregistered
[ 3259.811645] THROTTLER: device file closed
[ 3259.840103] THROTTLER: device file successfully opened for object with minor 0
[ 3259.840112] THROTTLER: somebody called an ioctl on dev with [major,minor] number [241,0], command 1074820105 and paramet
er 140725523064512
[ 3259.840116] THROTTLER: program test_syscall_by deregistered
[ 3259.840192] THROTTLER: device file closed
[ 3264.724262] THROTTLER: device file successfully opened for object with minor 0
[ 3264.724285] THROTTLER: somebody called an ioctl on dev with [major,minor] number [241,0], command 2183164938 and paramet
er 140735711205808
[ 3264.724505] THROTTLER: device file closed
```

D5 — Monitor off

`sudo ./user/throttlectl add-prog test_syscall_by_nr; sudo ./user/throttlectl off` poi `./test_syscall_by_nr 39 1`

Atteso: nessuna osservazione, nessun limite applicato (tutto istantaneo).

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl show
monitor: off
default MAX: unlimited
syscall registered: 39
EUID registered:
registered programs: test_syscall_by
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/test_syscall_by_nr 39 1
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo dmesg | tail -10
[ 3378.117306] THROTTLER: device file successfully opened for object with minor 0
[ 3378.117317] THROTTLER: somebody called an ioctl on dev with [major,minor] number [241,0], command 1074820104 and paramet
er 140720641310400
[ 3378.117480] THROTTLER: device file closed
[ 3378.136498] THROTTLER: device file successfully opened for object with minor 0
[ 3378.136509] THROTTLER: somebody called an ioctl on dev with [major,minor] number [241,0], command 29701 and parameter 0
[ 3378.136512] THROTTLER: monitor off
[ 3378.136604] THROTTLER: device file closed
[ 3384.728042] THROTTLER: device file successfully opened for object with minor 0
[ 3384.728067] THROTTLER: somebody called an ioctl on dev with [major,minor] number [241,0], command 2183164938 and paramet
er 140727640460320
[ 3384.728320] THROTTLER: device file closed
```

E — Throttling

Setting: `sudo make unmount && sudo make mount && sudo ./user/throttlectl on && sudo ./user/throttlectl add-sys 39 && sudo ./user/throttlectl add-prog test_syscall_by_nr`

E1 — Sotto MAX

`sudo ./user/throttlectl set-max 100` poi `time ./test_syscall_by_nr 39 50`

Atteso: tutte passano, nessun blocco, tempo ~istantaneo. `./user/throttlectl stats`: peak delay 0 ms

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl set-max 100
set-max: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ time ./user/test_syscall_by_nr 39 50

real    0m0.011s
user    0m0.001s
sys     0m0.006s
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl stats

peak delay: 0 ms
peak blocked threads: 0
average blocked threads: 0.000
```

E2 — Oltre MAX

`sudo ./user/throttlectl set-max 5` poi `time ./test_syscall_by_nr 39 30`

Atteso: rallentamento misurabile (secondi); `dmesg` `OVER MAX`; `stats` → peak delay > 0, program `test_syscall_by`, euid 1000, programma/euid responsabili corretti.

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl set-max 5
set-max: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ time ./user/test_syscall_by_nr 39 30

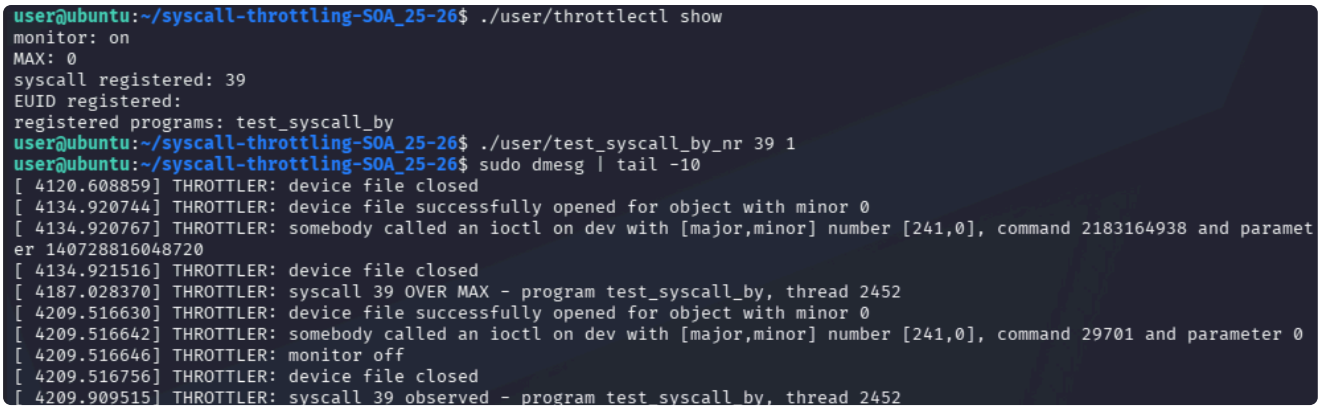
real    0m5.256s
user    0m0.004s
sys     0m0.004s
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo dmesg | tail -10
[ 3766.855531] THROTTLER: syscall 39 observed - program test_syscall_by, thread 2435
[ 3766.855541] THROTTLER: syscall 39 observed - program test_syscall_by, thread 2435
[ 3766.855549] THROTTLER: syscall 39 observed - program test_syscall_by, thread 2435
[ 3766.855556] THROTTLER: syscall 39 observed - program test_syscall_by, thread 2435
[ 3766.855563] THROTTLER: syscall 39 OVER MAX - program test_syscall_by, thread 2435
[ 3767.899014] THROTTLER: syscall 39 observed - program test_syscall_by, thread 2435
[ 3767.899048] THROTTLER: syscall 39 observed - program test_syscall_by, thread 2435
[ 3767.899057] THROTTLER: syscall 39 observed - program test_syscall_by, thread 2435
[ 3767.899064] THROTTLER: syscall 39 observed - program test_syscall_by, thread 2435
[ 3767.899071] THROTTLER: syscall 39 observed - program test_syscall_by, thread 2435
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl stats

peak delay: 1073 ms - program test_syscall_by, euid 1000
peak blocked threads: 1
average blocked threads: 0.021
```

E3 — MAX = 0 blocca tutto

term.1: `sudo ./user/throttlectl set-max 0` poi `./test_syscall_by_nr 39 1` (si blocca) · term.2: `sudo ./user/throttlectl off`

Atteso: la chiamata si **blocca** (nessuna invocazione ammessa). Verifica il rilascio con `off` da un altro terminale → la chiamata riparte. Atteso: con MAX=0 ogni chiamata registrata è bloccata; non si sblocca da sola.



E5 — Indipendenza per soggetto

registra prog A (`sudo ./user/throttlectl add-prog test_args`) **e** prog B (`sudo ./user/throttlectl add-prog test_syscall_by_nr`)

Atteso: il carico pesante su A non deve rallentare B (contatori separati).

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl show
monitor: on
MAX: 5
syscall registered: 39
EUID registered:
registered programs: test_syscall_by test_args
user@ubuntu:~/syscall-throttling-SOA_25-26$ time ./user/test_syscall_by_nr 39 20

real    0m3.104s
user    0m0.001s
sys     0m0.006s
user@ubuntu:~/syscall-throttling-SOA_25-26$ time ./user/test_syscall_by_nr 39 20

real    0m3.078s
user    0m0.000s
sys     0m0.008s
```

I primi valori sono durante l'esecuzione in parallelo, i secondo con l'esecuzione singola. Non ci sono interferenze.

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ time ./user/test_blockingsys 10

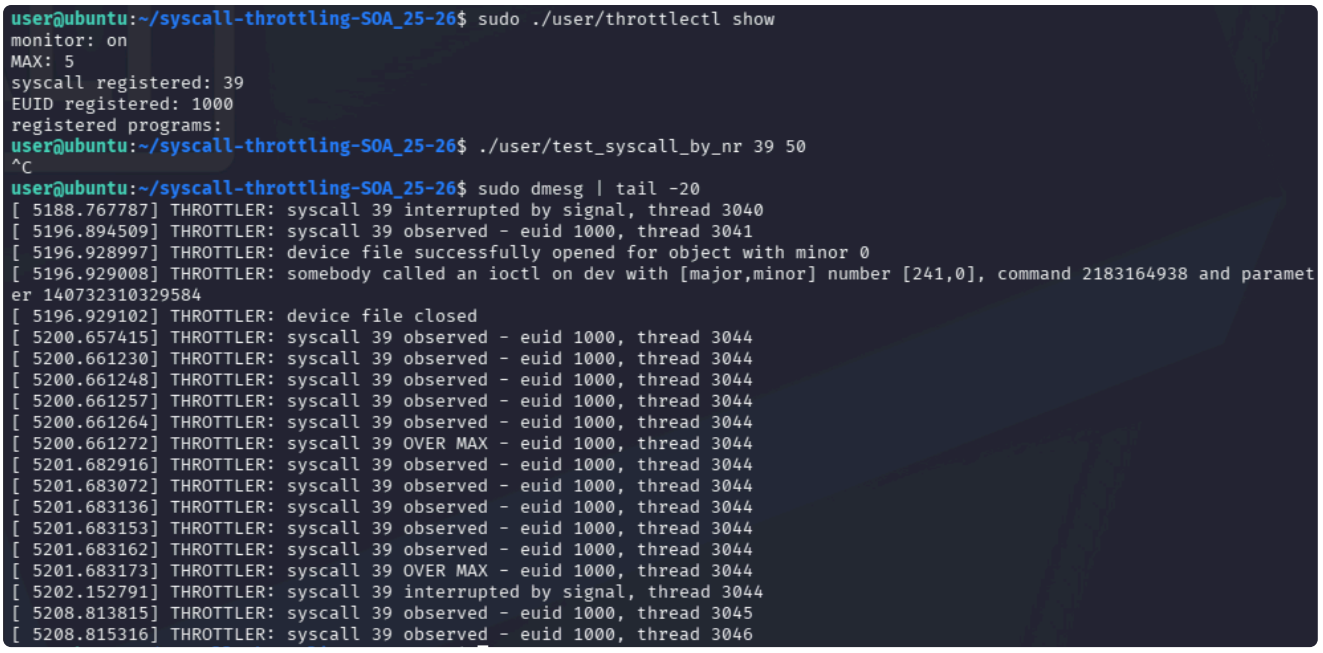
real    0m10.053s
user    0m0.000s
sys     0m0.010s
user@ubuntu:~/syscall-throttling-SOA_25-26$ time ./user/test_blockingsys 10

real    0m10.193s
user    0m0.000s
sys     0m0.006s
```

E6 — Segnale durante il blocco

`sudo ./user/throttlectl on; sudo ./user/throttlectl set-max 5` poi `./test_syscall_by_nr 39 50` e **Ctrl-C** mentre è bloccato

Atteso: il processo termina subito; `dmesg` conferma uscita su segnale `syscall 39 interrupted by signal, thread XXX`

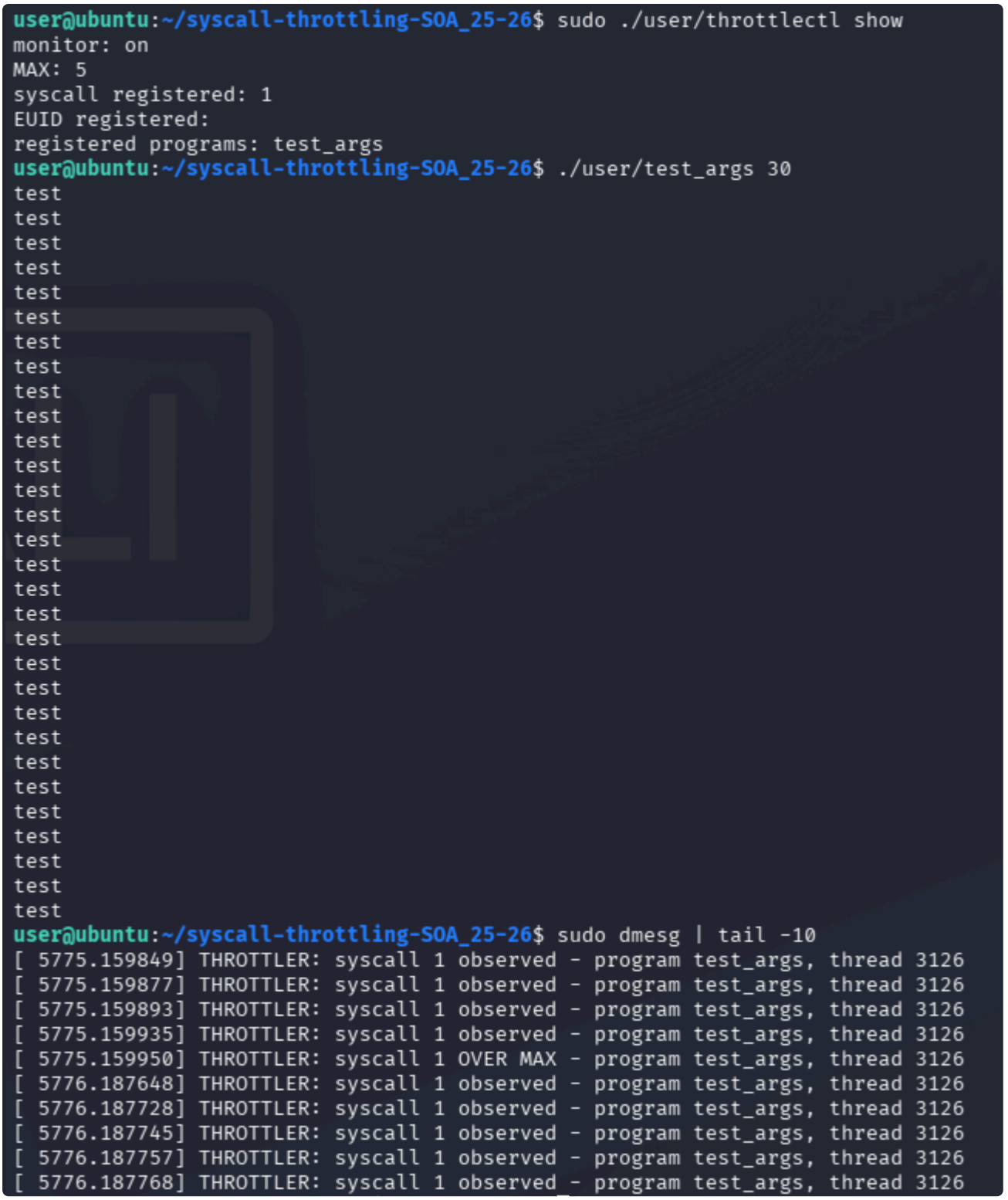


E7 — Syscall con argomenti (`write`, nr 1)

`add-sys 1`, `add-prog test_args`, MAX basso, `./test_args 30`.

Atteso: esattamente 30 `X` stampate (argomenti `fd/buf/count` passati intatti a `write` tramite `regs`) ma a ritmo rallentato dal throttle.

Registrato per **programma**, non euid, per non rallentare la shell. Dimostra che il wrapper non altera gli argomenti.



E8 — Syscall bloccante e con argomenti (`clock_nanosleep` , nr 230)

`sudo ./user/throttlectl add-sys 230; sudo ./user/throttlectl add-prog test_blockingsys; sudo ./user/throttlectl set-max 5` poi `time ./test_blockingsys 10`

Atteso: gira ≥10 s (i `sleep` funzionano) + attesa del throttle; `dmesg` `observed` / `OVER MAX` su 230 . Dimostra che una syscall bloccante con argomenti è gestita correttamente (il throttle e il blocco interno coesistono).

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl set-max 2
set-max: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ time ./user/test_blockingsys 10

real    0m4.272s
user    0m0.001s
sys     0m0.007s
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo ./user/throttlectl set-max 50
set-max: ok
user@ubuntu:~/syscall-throttling-SOA_25-26$ time ./user/test_blockingsys 10

real    0m0.547s
user    0m0.000s
sys     0m0.007s
```

F — Riconfigurazione a caldo (thread già bloccati)

Setting: `sudo make unmount && sudo make mount && sudo ./user/throttlectl on && sudo ./user/throttlectl add-sys 39 && sudo ./user/throttlectl add-prog test_blocked && sudo ./user/throttlectl set-max 5`

F1 — Monitor off mentre bloccato

Manda in blocco `./test_blocked 39 10 20` (MAX=5), poi da altro terminale monitor `off`

Atteso: i thread bloccati ripartono entro ~1 finestra; il test finisce; Nessun deadlock.

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl show
monitor: on
MAX: 5
syscall registered: 39
EUID registered:
registered programs: test_blocked
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/test_blocked 39 10 20
done: 10 processes x 20 calls on syscall 39
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo dmesg | tail -20
[ 2010.204631] THROTTLER: syscall 39 observed - program test_blocked, thread 1429
[ 2010.204633] THROTTLER: syscall 39 observed - program test_blocked, thread 1429
[ 2011.228268] THROTTLER: syscall 39 observed - program test_blocked, thread 1424
[ 2011.228268] THROTTLER: syscall 39 observed - program test_blocked, thread 1425
[ 2011.228285] THROTTLER: syscall 39 observed - program test_blocked, thread 1424
[ 2011.228285] THROTTLER: syscall 39 observed - program test_blocked, thread 1425
[ 2011.228288] THROTTLER: syscall 39 observed - program test_blocked, thread 1424
[ 2011.228309] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1424
[ 2011.228310] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1425
[ 2011.933705] THROTTLER: device file successfully opened for object with minor 0
[ 2011.933717] THROTTLER: somebody called an ioctl on dev with [major,minor] number [241,0], command 29701 and parameter 0
[ 2011.933721] THROTTLER: monitor off
[ 2011.933787] THROTTLER: device file closed
[ 2012.252385] THROTTLER: syscall 39 observed - program test_blocked, thread 1427
[ 2012.252584] THROTTLER: syscall 39 observed - program test_blocked, thread 1424
[ 2012.252743] THROTTLER: syscall 39 observed - program test_blocked, thread 1420
[ 2012.256785] THROTTLER: syscall 39 observed - program test_blocked, thread 1423
[ 2012.257461] THROTTLER: syscall 39 observed - program test_blocked, thread 1428
[ 2012.257647] THROTTLER: syscall 39 observed - program test_blocked, thread 1426
[ 2012.257776] THROTTLER: syscall 39 observed - program test_blocked, thread 1425
```

Le righe `observed` dopo `monitor off` sono i thread **già dentro** il wrapper che finiscono, non chiamate nuove.

F2 — Deregistrazione mentre bloccato

Stesso scenario di F1, poi `del-prog test_blocked`

Atteso: i thread ripartono entro ~1 finestra.

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/test_blocked 39 10 20
done: 10 processes x 20 calls on syscall 39
user@ubuntu:~/syscall-throttling-SOA_25-26$ sudo dmesg | tail -20
[ 3006.564533] THROTTLER: syscall 39 observed - program test_blocked, thread 1500
[ 3006.564535] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1504
[ 3006.564543] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1500
[ 3007.588522] THROTTLER: syscall 39 observed - program test_blocked, thread 1495
[ 3007.588560] THROTTLER: syscall 39 observed - program test_blocked, thread 1495
[ 3007.588567] THROTTLER: syscall 39 observed - program test_blocked, thread 1497
[ 3007.588571] THROTTLER: syscall 39 observed - program test_blocked, thread 1495
[ 3007.588580] THROTTLER: syscall 39 observed - program test_blocked, thread 1495
[ 3007.588591] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1495
[ 3007.588593] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1497
[ 3008.612564] THROTTLER: syscall 39 observed - program test_blocked, thread 1502
[ 3008.612595] THROTTLER: syscall 39 observed - program test_blocked, thread 1502
[ 3008.612605] THROTTLER: syscall 39 observed - program test_blocked, thread 1502
[ 3008.612612] THROTTLER: syscall 39 observed - program test_blocked, thread 1502
[ 3008.612619] THROTTLER: syscall 39 observed - program test_blocked, thread 1502
[ 3008.612627] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1502
[ 3009.489764] THROTTLER: device file successfully opened for object with minor 0
[ 3009.489777] THROTTLER: somebody called an ioctl on dev with [major,minor] number [241,0], command 1074820105 and paramet
er 140731974994064
[ 3009.489790] THROTTLER: program test_blocked deregistered
[ 3009.489868] THROTTLER: device file closed
```

F3 — Cambio MAX mentre bloccato

Stesso scenario di F1, poi `set-max 1000`

Atteso: al ri-conteggio della finestra successiva passano. Nessun blocco permanente.

```
[ 3058.137149] THROTTLER: program test_blocked registered
[ 3058.137231] THROTTLER: device file closed
[ 3068.380282] THROTTLER: syscall 39 observed - program test_blocked, thread 1521
[ 3068.380322] THROTTLER: syscall 39 observed - program test_blocked, thread 1521
[ 3068.380334] THROTTLER: syscall 39 observed - program test_blocked, thread 1521
[ 3068.380344] THROTTLER: syscall 39 observed - program test_blocked, thread 1521
[ 3068.380446] THROTTLER: syscall 39 observed - program test_blocked, thread 1521
[ 3068.380458] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1521
[ 3068.381759] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1522
[ 3068.384095] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1524
[ 3068.384618] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1525
[ 3068.385173] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1523
[ 3068.386127] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1526
[ 3068.387038] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1527
[ 3068.387088] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1529
[ 3068.388148] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1530
[ 3068.388290] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1528
[ 3069.415667] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3069.415730] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3069.415737] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3069.415742] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3069.415747] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3069.415752] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1523
[ 3070.439915] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3070.439950] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3070.439959] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3070.439966] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3070.439973] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3070.439980] THROTTLER: syscall 39 OVER MAX - program test_blocked, thread 1523
[ 3070.881379] THROTTLER: device file successfully opened for object with minor 0
[ 3070.881403] THROTTLER: somebody called an ioctl on dev with [major,minor] number [241,0], command 29699 and parameter 10
00
[ 3070.881408] THROTTLER: MAX set to 1000
[ 3070.881520] THROTTLER: device file closed
[ 3071.463840] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3071.463873] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3071.463882] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3071.463889] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3071.463896] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3071.463903] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3071.463910] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
[ 3071.463917] THROTTLER: syscall 39 observed - program test_blocked, thread 1523
```

G — Statistiche

Setting: `sudo make unmount && sudo make mount && sudo ./user/throttlectl on && sudo ./user/throttlectl add-sys 39 && sudo ./user/throttlectl add-prog test_blocked && sudo ./user/throttlectl set-max 5`

G1 — Azzeramento a freddo

Subito dopo `mount`, `stats`

Atteso: peak delay 0, peak blocked threads 0, average 0.000.

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl show
monitor: on
MAX: 5
syscall registered: 39
EUID registered:
registered programs: test_blocked
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl stats

peak delay: 0 ms
peak blocked threads: 0
average blocked threads: 0.000
```

G2 — Picco di ritardo + responsabili

Carico noto `./test_blocked 39 1 30` poi `./user/throttlectl stats`

Atteso: mostra peak delay plausibile, con programma ed euid corretti del chiamante responsabile.

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/test_blocked 39 1 30
done: 1 processes x 30 calls on syscall 39
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl stats

peak delay: 1025 ms - program test_blocked, euid 1000
peak blocked threads: 1
average blocked threads: 0.039
```

G3 — Picco e media thread bloccati

`test_blocked 39 10 3` (MAX=5)

Atteso: `stats`: peak blocked threads > 1 (fino a ~10) e average blocked threads > 0 . Verificabile in diretta da un 2° terminale.

```
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/test_blocked 39 10 3
done: 10 processes x 3 calls on syscall 39
user@ubuntu:~/syscall-throttling-SOA_25-26$ ./user/throttlectl stats

peak delay: 5134 ms - program test_blocked, euid 1000
peak blocked threads: 9
average blocked threads: 0.183
```

G5 — Lettura in qualsiasi momento

`stats` funziona durante un carico attivo (non solo a riposo).
Atteso: il peak sale in diretta; `stats` risponde sempre Confermato

---

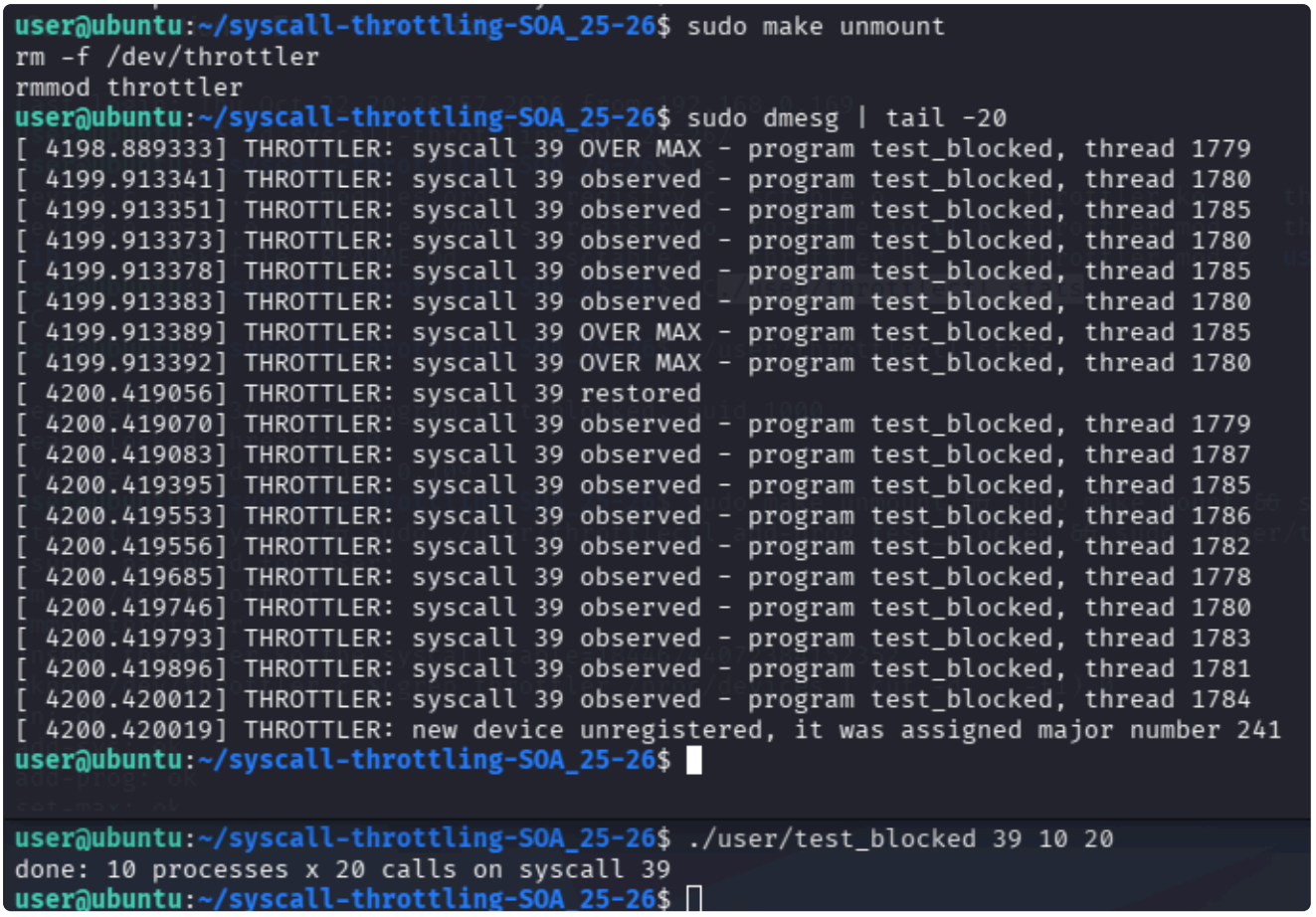
H — Rimozione sicura

Setting: `sudo make unmount && sudo make mount && sudo ./user/throttlectl on && sudo ./user/throttlectl add-sys 39 && sudo ./user/throttlectl add-prog test_blocked && sudo ./user/throttlectl set-max 5`

H1 — rmmod con thread bloccato

Manda in blocco un thread, term.1: `./test_blocked 39 10 20` · term.2: `sudo make unmount`

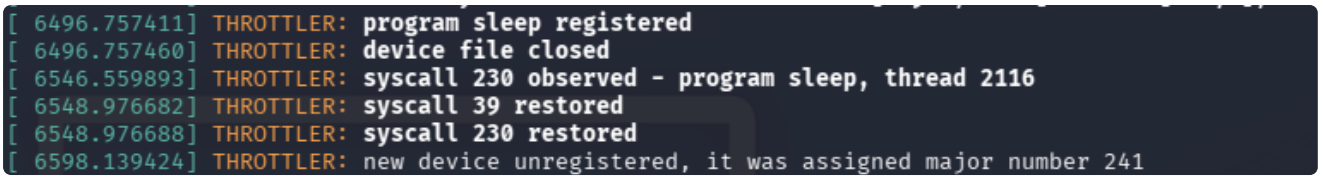
Atteso: il thread viene rilasciato, la syscall completa, `dmesg` : tabella ripristinata, device deregistrato, **nessun crash/hang**.



H2 — Unmount con thread dentro la syscall bloccante reale

Soggetto registrato su una syscall bloccante a tempo **finito**, poi `rmmod` mentre un thread è dentro la syscall originale.
`sudo ./user/throttlectl add-sys 230; sudo ./user/throttlectl add-prog sleep`.
term.1: `sleep 5` · term.2 (entro i 5 s): `sudo make unmount`

Atteso: il drain **attende** il ritorno di `clock_nanosleep` (~5 s), poi completa; `dmesg` : `observed` su 230 poi device deregistrato; **nessun crash né panic**. Dimostra che un thread fermo nella syscall *reale* (non nella wait-queue del monitor) viene atteso in sicurezza. Il caso a blocco **indefinito** (es. `poll` di sistema) tiene lo scarico in attesa.



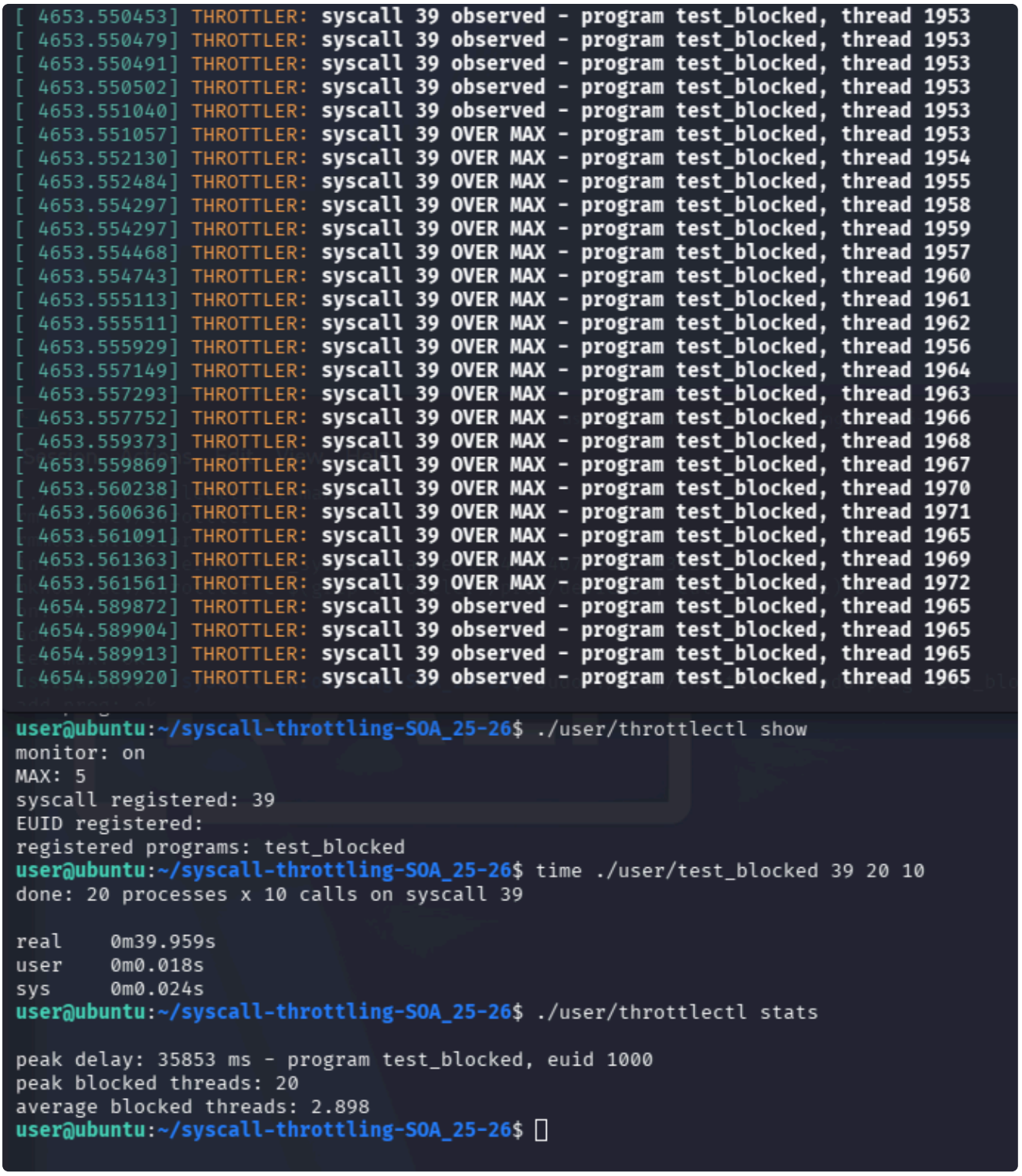
I — Concorrenza

Setting: `sudo make unmount && sudo make mount && sudo ./user/throttlectl on && sudo ./user/throttlectl add-sys 39 && sudo ./user/throttlectl set-max 5`

I1 — Molti processi concorrenti

`sudo ./user/throttlectl add-prog test_blocked` poi `./test_blocked 39 20 10`

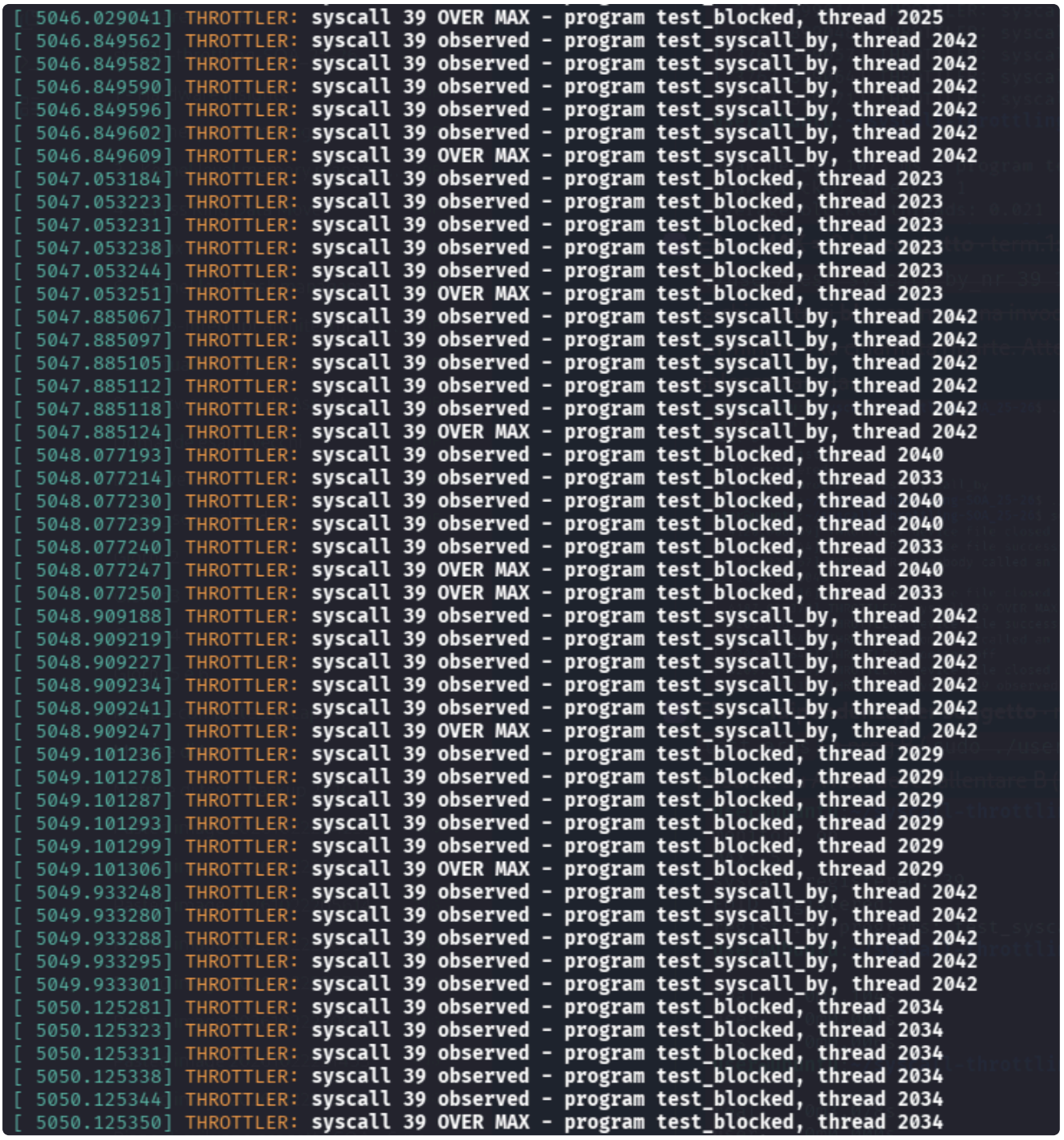
Atteso: nessun crash, statistiche coerenti, `dmesg` ordinato.
(più thread che condividono lo stesso prog-name vengono bloccati insieme, il blocco è per prog o per euid)



I2 — Più syscall registrate insieme

registra 39 + altre 2-3 syscall non bloccanti, `sudo ./user/throttlectl add-sys 1; sudo ./user/throttlectl add-sys 230` poi genera carico misto (`test_args` , `test_blockingsys`)

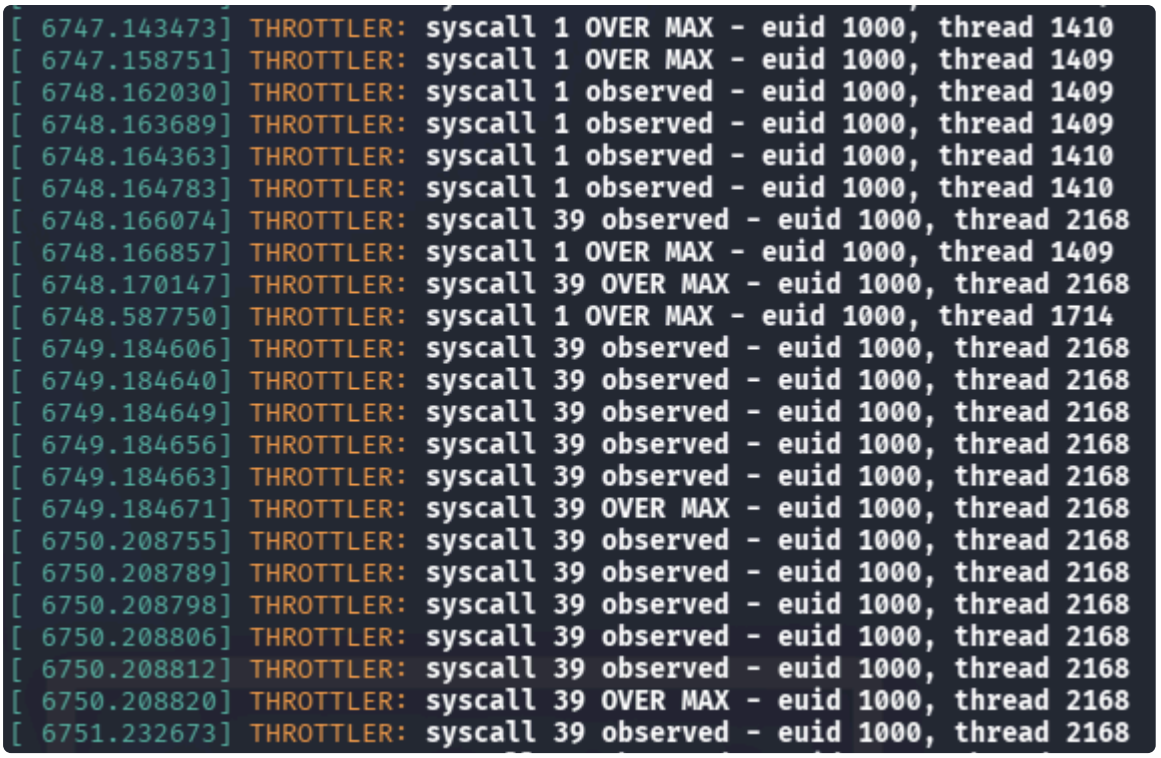
Atteso: ognuna osservata/limitata correttamente.



I3 — Un soggetto, due syscall: budget condiviso

Un unico soggetto (EUID) con **due** syscall registrate : il suo contatore è condiviso tra le due (il `count` è per-soggetto, non per-syscall).
Setting: `on; add-sys 39; add-sys 1; add-uid 1000; set-max 5` . Lanciare insieme, come utente 1000, `./test_args 20` (write) e `./test_syscall_by_nr 39 20` (getpid)

Atteso: in `dmesg` i 5 posti/finestra dell'EUID 1000 sono spartiti **tra `write` e `getpid` insieme** (~5 totali per finestra, non 5+5).



Programmi di test usati

- `test_ioctl` — esercita ogni comando ioctl e i relativi percorsi d'errore (doppia registrazione, deregistrazione di una voce assente, comando sconosciuto, syscall fuori intervallo e non-returning); eseguito con `sudo` per i percorsi d'errore, senza `sudo` per verificare i permessi.
- `test_syscall_by_nr <nr> <n>` — esegue un processo con `n` invocazioni della syscall `nr`.
- `test_args <n>` — lancia `n` chiamate `write` (syscall con argomenti): per verificare che gli argomenti siano inoltrati intatti.
- `test_blockingsys <n>` — esegue `n` chiamate `clock_nanosleep` (syscall bloccante): verifica che una syscall bloccante sia limitata mentre il suo blocco interno continua a funzionare.
- `test_blocked <nr> <nproc> <calls>` — lancia `nproc` processi che chiamano la syscall `nr` in concorrenza sullo stesso soggetto (statistica picco/media thread bloccati, concorrenza).